PROJECT SYNOPSIS

Group ID : 31

Project Title : IoT-Enabled Tile for Controlled Substance Inventory Tracking and Access Verification

1. Project Definition

Hospitals and healthcare facilities are required to maintain strict accountability over controlled substances such as scheduled drugs, narcotics, and other high-value or restricted medications. Existing practices in many facilities still rely on manual logbooks, periodic physical counting, or basic barcode-based tracking, all of which are vulnerable to human error, delayed detection of discrepancies, and, in some cases, deliberate tampering or diversion of stock. Any gap in the chain of custody of a controlled substance can have serious legal and patient-safety consequences, and existing methods provide no real-time visibility into who accessed a substance, when, and how much was taken.

This project proposes a low-cost, sensor-fusion-based smart storage tile that automates the tracking of controlled substances at the point of storage. The system combines multi-point load-cell weight sensing, dual RFID verification (of both the product and the accessing staff member), an infrared motion sensor, a camera module for visual evidence, and an environmental sensor, to build a reliable, auditable record of every item added to or removed from the tile. All sensor data is processed by a backend server that manages staff access sessions, calculates item counts, applies configurable anomaly-detection rules, and raises real-time alerts through a web dashboard and a messaging bot. The result is a practical, demonstrable system that strengthens accountability for controlled-substance storage.

2. Scope

- Design and development of a single smart storage tile integrating multi-point load-cell weight sensing, dual RFID identification, IR motion sensing, a camera module, and an environmental sensor.
- Implementation of a session-based access control mechanism that attributes every item-level transaction to an authenticated staff member.
- Development of a web-based dashboard for administrators and staff to monitor inventory, manage users and products, and review transaction/alert history.
- Real-time alert notification to designated personnel through a messaging platform.
- Secure logging of all inventory transactions with user identification, timestamps, sensor readings, and photographic evidence

3. Objectives

1. To design a smart storage tile that automatically determines item count using multi-point load-cell weight sensing.
2. To implement dual RFID verification for identifying both the product/container and the staff member accessing the tile.
3. To implement a session-based access model that attributes all item-level transactions occurring during a session to the authenticated user.
4. To detect and flag anomalous access events, such as withdrawal of more units than permitted per session or an identity mismatch.
5. To monitor environmental storage conditions (temperature and humidity) and raise alerts when safe thresholds are breached.

6. To build a web-based dashboard and a real-time notification system for transaction visibility, alert review, and user/product management.

4. Review of Conference/Journal Papers and Relevant Theory

A review of existing academic and industry work relevant to this project reveals several related but distinct lines of work, summarized below.

4.1 RFID-based medication and dispenser tracking

Several studies on IoT-enabled medical dispensers use RFID to verify patient or medication identity before a dose is released, and pair this with a load cell to confirm whether the intended quantity was actually dispensed, reducing missed or incorrect doses. Some of these systems also integrate microcontroller-based alert mechanisms to remind patients or caregivers of dosage timings, combining identity verification with scheduling logic. These systems demonstrate that combining RFID identity checks with weight-based confirmation is a practically validated approach for medication accountability, though this line of work is largely centered on the dispensing of doses to individual patients rather than on securing bulk shelf-level or cabinet-level stock used by staff across an entire ward.

4.2 RFID and sensor-based hospital asset and specimen tracking

Broader hospital IoT research proposes wireless sensor network architectures combining RFID with low-power wireless protocols to track staff, patients, and biomedical equipment in real time, feeding data to a central monitoring system accessible over a REST-based web service. Such architectures typically separate the system into a device/tag layer, a reader/gateway layer, and a server layer, which is conceptually similar to the layered structure proposed in this project. Related work on hospital specimen logistics uses RFID tags on sample tubes together with environmental (temperature/humidity) sensors on transport containers to ensure specimens are not lost, misdirected, or exposed to unsafe conditions in transit. These works validate the use of RFID plus environmental sensing as a reliable combination for chain-of-custody problems in a hospital setting, which is conceptually close to the requirement of this project, though applied to specimens in transit rather than stationary controlled-substance stock.

4.3 Weight-based smart shelf systems (retail and industrial)

Outside healthcare, smart shelf systems are an established product category in retail and warehouse inventory management, typically combining load cells with RFID tags to continuously estimate stock counts without requiring manual scanning. Patented inventory-cabinet designs describe a comparable approach: load cells embedded in shelves determine the total weight change on a shelf, which is converted into an item count using a known reference weight per unit, while an RFID reader identifies the individual responsible for the transaction. Several of these designs also discuss compensating for sensor drift and calibration error over repeated use, a practical concern that is directly relevant to the weight-to-count logic used in this project. This closely mirrors the core weight-to-count and identity-attribution logic proposed here, though existing systems in this space are generally designed for retail stock-level visibility rather than for generating a security-oriented, tamper-aware audit trail suited to regulated substances.

4.4 Security considerations in IoT-based medical supply chains

Research on RFID security in medical and pharmaceutical supply chains highlights that passive RFID tags, being resource-constrained, are susceptible to a range of security threats including cloning, eavesdropping, and relay attacks, and argues for security to be considered at every layer of an IoT deployment used for tracking high-value or regulated drugs. This body of work recommends layered verification and anomaly monitoring rather than relying on a single authentication factor, which directly motivates the inclusion of multi-sensor verification (RFID, weight, and motion sensing together) in the present project rather than RFID identification alone.

4.5 Gap identified

While RFID-based dispensers, hospital asset-tracking systems, and retail smart shelves each individually address parts of this problem, no reviewed system combines multi-corner load-cell weight sensing, dual RFID verification (for both the substance and the accessing staff member), presence/motion confirmation, environmental monitoring, and photographic evidence capture into a single point-of-storage device purpose-built for controlled-substance

accountability in a clinical setting. Existing dispenser systems focus on patient-level dosing, hospital asset-tracking systems focus on specimens or equipment in transit, and retail smart shelves focus on stock visibility rather than tamper-aware auditability.The proposed system combines multi-point weight sensing, dual RFID verification, IR motion detection, environmental monitoring, and photographic evidence into a unified storage-monitoring solution for controlled substances.

5. System Architecture

The proposed system follows a layered architecture consisting of a sensing layer, a communication layer, an application layer, a data layer, a presentation layer, and a notification layer, as illustrated below.

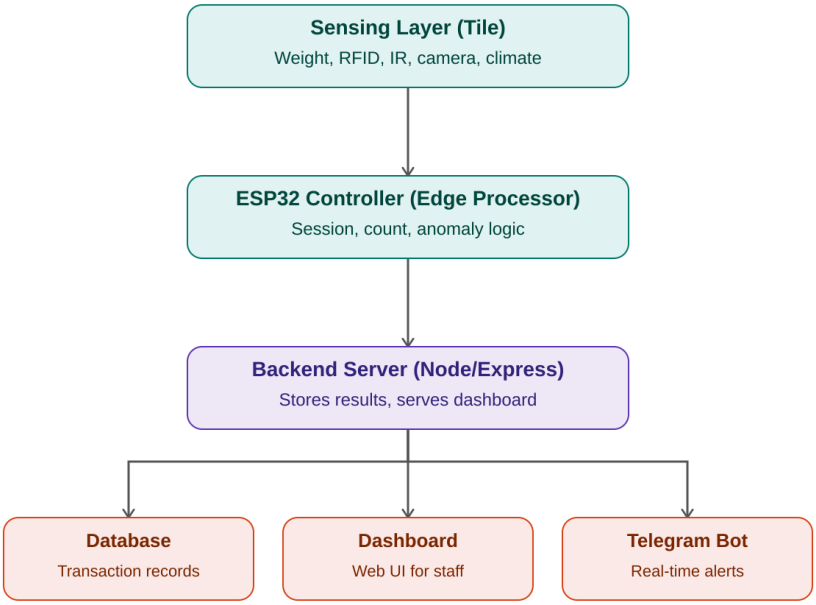


Figure 1: System Architecture

Sensor data from the tile (weight, RFID, IR, camera, temperature/humidity) is read by the ESP32 microcontroller and transmitted to the backend server over HTTP. All transaction logic — session tracking, item-count calculation, threshold-based anomaly detection, and alert generation — is handled centrally on the backend rather than on the microcontroller itself, so that business rules such as per-product withdrawal limits can be updated without modifying the embedded firmware. Verified transactions, user and product records, and alert history are stored in a database and made available through a web-based dashboard, while critical alerts are additionally pushed to a messaging bot for immediate notification to relevant staff.

The access-session logic that governs when a transaction is attributed to a staff member is shown in the flow below.

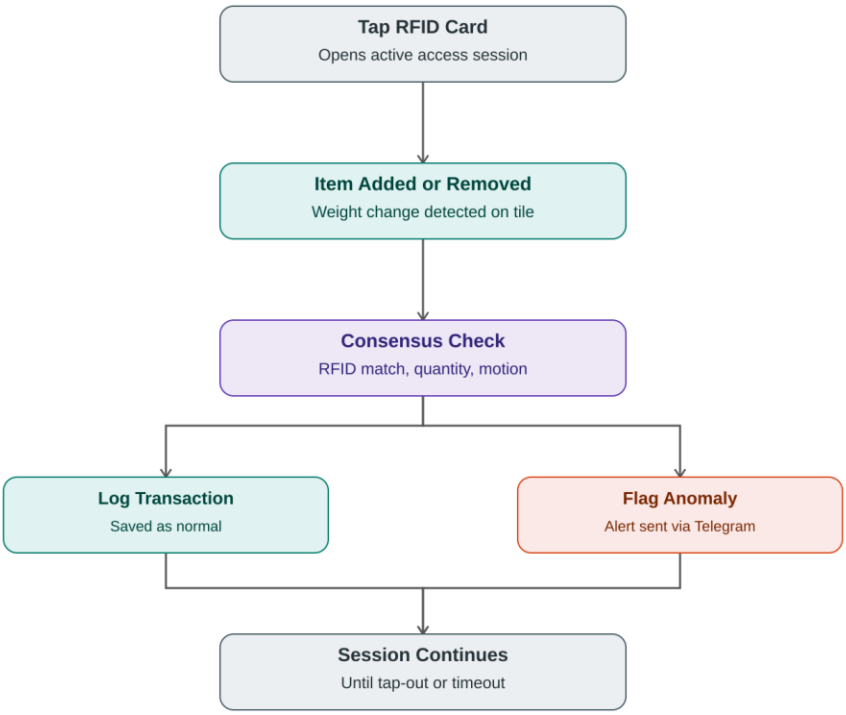


Figure 2: Access Session and Transaction Flow

A staff member begins an access session by tapping their RFID card at the tile. Any item added or removed while the session is active is checked against three signals together — RFID identity, weight-based quantity change, and IR-confirmed physical motion — before being logged as a normal transaction. If these signals do not agree, or if the quantity removed exceeds the configured limit for that product, the event is flagged as an anomaly and an alert is raised. The session remains active until the staff member taps out or an inactivity timeout is reached, whichever occurs first.

6. System Requirements

Hardware Requirements

- ESP32 microcontroller (central controller for the tile)
- 4 × Load cells with HX711 load cell amplifier module
- 2 × RFID reader modules with tags/cards (product identification and staff identification)
- 1 × IR break-beam / proximity sensor
- 1 × ESP32-CAM module (photographic evidence capture)
- 1 × Temperature and humidity sensor (BME280)
- Development machine — Processor: Intel Core i3 / Ryzen 3 or higher; RAM: 4 GB minimum (8 GB recommended); Hard Disk: 256 GB

Software Requirements

- Operating System: Windows 10/11 or Linux (development machine)
- Browser: Chrome, Firefox, or Edge (for dashboard access)

- Firmware: Arduino IDE / ESP-IDF framework (C/C++) for sensor interfacing and HTTP communication
- Backend: Node.js with Express.js
- Database: SQL
- Frontend: React.js
- Notifications: Telegram Bot API
- Version Control: Git and GitHub

7. Conclusion

Manual and partially digitized methods of tracking controlled substances in healthcare facilities leave real gaps in accountability, with no real-time visibility into stock levels, no automated verification of staff access, and no mechanism to detect unusual withdrawal patterns before they become a discrepancy. This project addresses these gaps by proposing a smart storage tile that combines weight sensing, dual RFID verification, motion confirmation, environmental monitoring, and photographic evidence capture into a single auditable system, with all decision logic handled by a backend server and surfaced through a web dashboard and real-time messaging alerts.

At the conclusion of this project, a fully functional single-tile prototype will be delivered, capable of automatically counting stored items, verifying both product and staff identity, detecting and flagging anomalous access, monitoring storage conditions, and maintaining a complete transaction log with visual evidence. While the current prototype is demonstrated for controlled-substance tracking, the same underlying design — weight-based verification combined with layered identity checks — is intended to generalize to other high-value, regulated inventory such as jewellery or IT assets with minor adjustments. The project is expected to demonstrate a practical, low-cost approach to strengthening accountability for controlled substance storage in resource-constrained healthcare settings.

8. References

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Group Members :-

Sr. No.	Roll No	Student Name	Signature
1	78	Laghave Harsh Sandesh	
2	88	Mundhe Aditya Vijay	
3	89	Muntode Simran Ravi	
4	137	Zalte Harshwardhan	

Prof. P.B Dhanawate **(Project Guide)**	**Dr. S.R Deshmukh** **(Project Coordinator)**	**Dr. M.A Jawale** **(HOD Computer Department)**